

UNIT – 3 : Proxy Servers, Tor & Privacy Protection Notes

1. Proxy Server

A proxy server acts as an intermediary between a user and the internet. It receives requests from the user and forwards them to websites, hiding the user's IP address.

Working:

User → Proxy Server → Website → Proxy Server → User

Advantages: Hides IP address Improves privacy Access blocked content Filters websites

Disadvantages: May slow browsing Some proxies log activity Free proxies may be unsafe **2.**

Types of Proxies

HTTP Proxy Handles normal web traffic. Used for webpage browsing and filtering. Fast but not

secure. **HTTPS Proxy** Handles encrypted web traffic. Used in banking, shopping and logins.

Provides secure communication. **SOCKS Proxy** Supports web, email, gaming and file transfer.

Flexible and supports multiple protocols. Commonly used with VPNs and Tor. **Comparison**

HTTP: Fast, Low Security

HTTPS: Secure, Encrypted

SOCKS: Flexible, Supports All Traffic

3. Browser Proxy Configuration Open browser settings. Navigate to Network/Proxy Settings.

Enter proxy IP address and port. Save and restart browser. **4. Anonymous Browsing and Tor**

Tor (The Onion Router) provides anonymous browsing using multiple layers of encryption. Routes traffic through multiple relays. Hides user identity and location. Makes tracking difficult. **5. IP**

Address and Masking

An IP address uniquely identifies a device on a network. Protects privacy when masked. Prevents tracking. Can bypass geographic restrictions. Methods: Proxy Server Tor Browser VPN **6.**

Privacy-Focused Browsers Block trackers. Prevent fingerprinting. Force secure HTTPS connections. Automatically clear browsing data. **7. Firefox Proxy Configuration** No Proxy – Direct connection. Auto Detect Proxy – Uses WPAD. Manual Proxy – User enters IP and port. Automatic Proxy URL – Loads proxy settings automatically. **8. FoxyProxy**

FoxyProxy is a Firefox extension used to manage multiple proxy servers. One-click proxy switching. Stores multiple proxy configurations. Website-based proxy rules. Supports HTTP, HTTPS, SOCKS4 and SOCKS5. Import and export proxy settings. **Conclusion**

Proxy servers, Tor, VPNs and privacy-focused browsers help protect user privacy, hide IP addresses and improve online security. Understanding these tools is essential for safe and secure internet usage.